FOUR EQUIVALENT DESCRIPTIONS OF R-UNIVERSALITY OF A SEMISIMPLE LIE ALGEBRA OVER Q

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Allowed standard prerequisites (editorial).

- The Tits algebras D(Lie G, K)(lambda) attached to a dominant weight and the representations rho_lambda over Q and over R. Set up in the author's introduction and in the definitions preceding the counted statement (source lines 79-201). Supplies the four objects that the four equivalent conditions speak about.

Lemma 0.1. *If $\mathfrak{g}$ is a semisimple Lie algebra over $\mathbb{Q}$, then the following are equivalent:*

1. *$\mathfrak{g}$ is $\mathbb{R}$-universal.*
2. *${}^{\mathbb{Q}}\rho_\lambda \otimes_{\mathbb{Q}} \mathbb{R} \cong {}^{\mathbb{R}}\rho_\lambda$, for every dominant weight λ.*
3. *$D_{\mathfrak{g},\mathbb{Q}}(\lambda) \otimes_{\mathbb{Q}} \mathbb{R}$ is a division algebra, for every dominant weight λ.*
4. *$D_{\mathfrak{g},\mathbb{Q}}(\lambda) \otimes_{\mathbb{Q}} \mathbb{R} \cong D_{\mathfrak{g},\mathbb{R}}(\lambda)$, for every dominant weight λ.*

Proof. (1 $\Rightarrow$ 2) Let ρ be a $\mathbb{Q}$-form of ${}^{\mathbb{R}}\rho_\lambda$. Then ρ must be $\mathbb{Q}$-irreducible (since $\rho \otimes_{\mathbb{Q}} \mathbb{R} \cong {}^{\mathbb{R}}\rho_\lambda$ is $\mathbb{R}$-irreducible) and

$$\rho \otimes_{\mathbb{Q}} \mathbb{C} \cong (\rho \otimes_{\mathbb{Q}} \mathbb{R}) \otimes_{\mathbb{R}} \mathbb{C} \cong {}^{\mathbb{R}}\rho_\lambda \otimes_{\mathbb{R}} \mathbb{C}$$

has an irreducible summand with highest weight λ. Therefore $\rho \cong {}^{\mathbb{Q}}\rho_\lambda$.

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$(2 \Rightarrow 1)$ ${}^{\mathbb{Q}}\rho_\lambda$ is a $\mathbb{Q}$-form of ${}^{\mathbb{R}}\rho_\lambda$.

$(2 \Leftrightarrow 3 \Leftrightarrow 4)$ We have $\operatorname{End}_{\mathfrak{g}}({}^{\mathbb{Q}}\rho_\lambda \otimes_{\mathbb{Q}} \mathbb{R}) = D_{\mathfrak{g},\mathbb{Q}}(\lambda) \otimes_{\mathbb{Q}} \mathbb{R}$. So ${}^{\mathbb{Q}}\rho_\lambda \otimes_{\mathbb{Q}} \mathbb{R}$ is irreducible (and hence equal to ${}^{\mathbb{R}}\rho_\lambda$) if and only if $D_{\mathfrak{g},\mathbb{Q}}(\lambda) \otimes_{\mathbb{Q}} \mathbb{R}$ is a division algebra (and hence equal to $D_{\mathfrak{g},\mathbb{R}}(\lambda)$). □

References

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[2] Prasad G., Rapinchuk A., On the existence of isotropic forms of semi-simple algebraic groups over number fields with prescribed local behavior, *Adv. Math.* **207** (2006), 646–660.

[3] Tits J., Représentations linéaires irréductibles d'un groupe réductif sur un corps quelconque, *J. Reine Angew. Math.* **247** (1971), 196–220.